Khush A. Patel

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Objective

Mechanical engineering student seeking a summer internship focused on sensor systems, real-time control, and mechanical–aerodynamic design. Interested in applying CAD, simulation, signal processing, and modeling to experimental hardware and sensing technologies.

Education

Johns Hopkins University, Whiting School of Engineering — Baltimore, MD
B.S. in Mechanical Engineering — Anticipated May 2029
Relevant Coursework: CAD and Design, Mechanics, Python, Digital Electronics, Electricity and Magnetism, Multivariable Calculus

Technical Skills

- **Software:** SolidWorks, ANSYS (Structural, Fluent, CFX), MATLAB, ROS 2, Webots, OnShape, Multisim
- **Programming/Analysis:** Python, MATLAB, C++, FEA, CFD, FFT, control systems, data visualization
- **Prototyping:** 3D printing, laser cutting, soldering, Arduino, mechanical assembly and testing

Engineering Experience

Amarkosha Wind Harvester Start-Up — Baltimore, MD
Mechanical Lead, Product Development — Jul 2025–Present

- Led **VIV wind harvester** design and multiphysics simulation in SolidWorks and ANSYS.
- Performed two-way transient **FSI analysis** (Fluent/CFX + Structural) to assess aerodynamic–structural coupling.
- Applied **FFT-based signal processing** in MATLAB to extract vortex shedding and lock-in behavior.
- Optimized geometry for wake effects, yielding **166% wind-speed amplification**.
- Accepted to **SPARK Accelerator Program** and selected as **BII Grant finalist**.

Johns Hopkins Mars Rover Team – Mechanical Subteam — Baltimore, MD
Mechanical & Controls Member (Wrist Mechanism) — Sep 2025–Present

- Designed a **3-DOF wrist** to enhance manipulator dexterity and payload handling.
- Conducted **torque analysis** for motor and joint optimization.
- Integrating **ROS 2** for real-time control and sensor feedback.

GreenWorks Development — Mechanicsburg, PA
Technical Sales & Project Intern — Jan–Aug 2025

- Modeled solar systems in Helioscope and Excel for a $5M performance validation project, contributing to proposal presentations with corporate leadership.

UPMC Magee–Women's Hospital — Pittsburgh, PA
Research Assistant, Dept. of Anesthesiology — Jun–Sep 2024

- Processed and visualized clinical datasets in R to support pre-eclampsia diagnostics research.

Projects

4-DOF Robotic Arm Simulation (Independent)

- Designed and simulated a 4-DOF manipulator in **SolidWorks** and **Webots** with **ROS 2** control.
- Integrated Arduino actuation and inverse kinematic feedback for closed-loop motion.
- Applied **Design for Manufacturing (DFM)** principles and **static FEA** to ensure mechanical reliability.

Leadership & Honors

- **President & Founder, Renewable Energy Club (RENEW)** — Led sustainable energy design projects and guest seminars.
- **Facilities Chair, Mini-THON** — Managed logistics and volunteers for $280K pediatric cancer fundraiser.
- **Honors:** Science Hall of Fame, William R. Pierce Physics Award, Lois Wolf GPA Athlete Award, Melissa Huang Scholarship.